higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICAL LITERACY

(Second Paper)

NQF LEVEL 2

(10401012)

25 February 2019 (X-Paper)
09:00–12:00

Calculators and drawing instruments may be used.

This question paper consists of 11 pages and 2 answer sheets.

NCV

TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start EACH question on a NEW page.
5. Show ALL the calculations clearly.
6. Round off ALL final answers appropriately according to the given context, unless stated otherwise.
7. Indicate units of measurement where applicable.
8. Diagrams are not necessarily drawn to scale.
9. Answer QUESTION 3.1.3 on ANSWER SHEET 1 (attached).
   Answer QUESTION 4.2.2 on ANSWER SHEET 2 (attached).
   Hand in both ANSWER SHEETS with your ANSWER BOOK.
10. Write neatly and legibly.

NCV

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QUESTION 1

1.1 A cattle farmer decided to practice rotational grazing. Rotational grazing is a system of moving cattle from one fenced piece of grazing land (paddock) to another to allow the land to recover and the grazing grass (pasture) to grow.

The diagram below shows how the farmer plans to divide the grazing land into three paddocks.
Paddock A and Paddock B have the same dimensions with a length of 240 metres and a breadth of 60 metres.

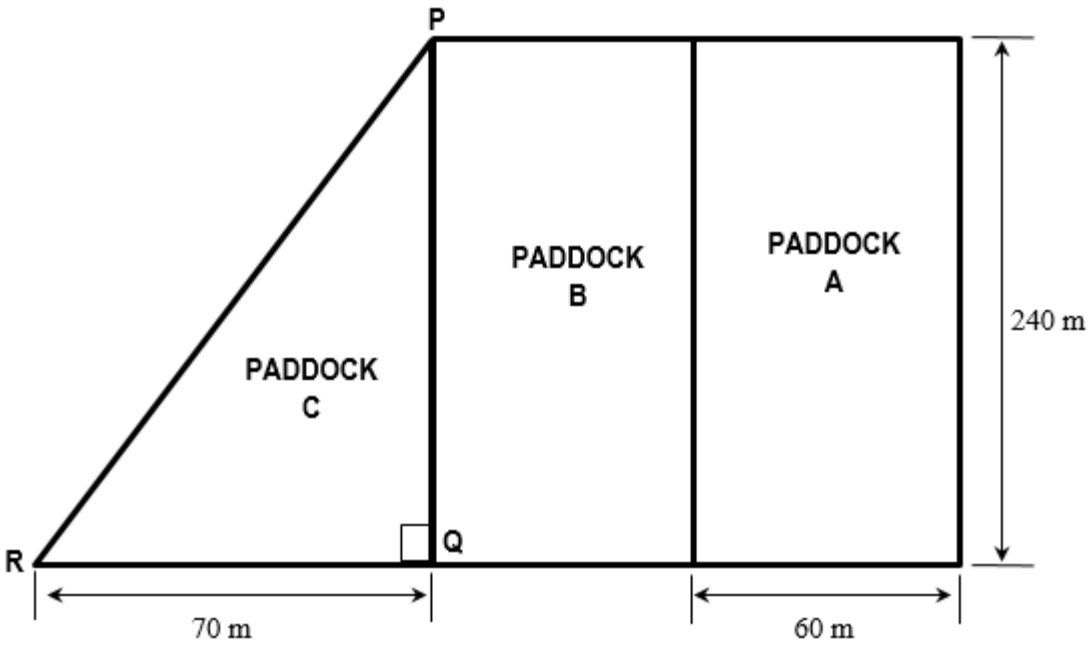


FIGURE 1

Study the diagram in FIGURE 1 above and answer the questions.

1.1.1 Name the shape represented by A. (1)

1.1.2 Name the shape represented by C. (1)

1.1.3 Calculate the total area of the grazing land.

Formulae: Area of rectangle = length × breadth

Area of triangle = $\frac{1}{2}$ × base × height (9)

1.1.4 Calculate the length of the hypotenuse PR.

Theorem of Pythagoras: $(PR)^2 = (PQ)^2 + (QR)^2$ (4)

1.1.5 Show that the total length of fencing required to fence the entire piece of grazing land is 1 280 m. (6)

1.1.6 Calculate the cost of fencing the entire grazing land if fencing costs R45 per metre. (3)

1.1.7 The map of the town on which the cattle is situated has a scale of 1 : 500 000 and the actual distance between the farm and the general store in town is 27 kilometres.

Determine the map distance between the farm and the general store in mm.

HINT: 1 km = 1 000 000 mm (6)

1.2 FIGURE 2 below shows the two types of milk bottles with which the farmer feeds the calves on the farm. Study the diagrams and answer the questions.





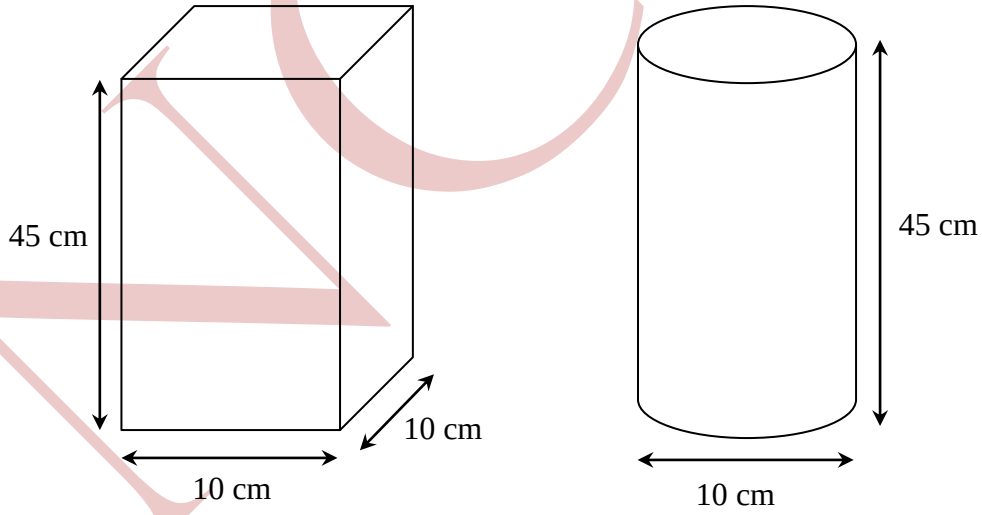


FIGURE 2

1.2.1 Calculate the volume of the rectangular shaped milk bottle.

Formula: Volume = length × breadth × height (5)

1.2.2 Calculate the volume of the cylindrical shaped milk bottle.

Formula: Volume = $\pi \times r^2 \times h$, where $\pi = 3,14$ (5)

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QUESTION 2

2.1 Kedibone recorded her income and expenses for the month of May 2017.

Study Kedibone's income and expenses below and answer the questions.

	May 2017
Total Income:	**R9 956,55**
Expenditure:	
Groceries	R1 800
Cell phone	R200
Car repayment	R1 300
Toiletries	R200
Car insurance	R440
School fees	R500
Entertainment	R300
Petrol	R430
Clothing account	R300
Medical aid	R1 225
Water and electricity	R400
Rent	R2 300
Credit card repayment	R150
Savings	R150
Total Expenditure:	**xxx**
Surplus/Deficit:	**R261,55**

2.1.1 Calculate Kedibone's total expenditure for May 2017. (2)

2.1.2 Show how the Surplus/Deficit value of R261,55 was calculated. Indicate whether this amount is a surplus or a deficit. (3)

2.1.3 Name TWO variable expenditures from the table. (2)

2.1.4 Name TWO fixed expenditures from the table. (2)

2.1.5 Do you think that Kedibone is in a good financial position? Justify your answer. (3)

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2.1.6 Advise Kedibone on what she could do with the money that is left over at the end of the month. Justify your answer. (2)

2.1.7 Calculate Kedibone's new surplus or deficit amount if her salary remained the same but the cost of her car insurance, medical aid and rent increased by 10%. Show ALL calculations. Indicate whether the amount is a surplus or a deficit (7)

2.2 Study John Smith's salary advice for the month of June 2017 below and answer the questions.

SALARY ADVICE			
Employee information			
Employee number **987**	**Name & Surname** **John Smith**	**ID nr** **9911050 0158 082**	**Tax number** **0740 214 17 7**
Pay date		Bank	
25/06/2017		**First Bank**	
Income		Deductions	
Description	**Amount**	**Description**	**Amount**
Basic Salary	**R7 998,00**	UIF	R88,73
Commission	**R875,00**	Medical Aid	R550,00
		PAYE	R2 218,25
		Pension Fund	R490,98
		Bond Repayment	R700,00
Gross salary	**R xxxxxxx**	Total Deductions	**R xxxxxxxx**
		Net Salary	**R xxxxxxx**

2.2.1 What is a salary advice? (2)

2.2.2 Calculate John's gross salary. (2)

2.2.3 Write the abbreviation PAYE in full. (1)

2.2.4 What percentage of John's gross salary does he pay towards PAYE? (3)

2.2.5 Calculate John's net salary. (4)

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2.3 John Smith repaired his car at Nutech Auto Repair.

Study the receipt John received as proof of payment below and answer the questions.

RECEIPT:

Date: 27 May 2017 No. 2234777

Received from: Tommy **Amount** R250,80

Amount: Two hundred and fifty rands and eighty cents only

For payment of: Wheel alignment and balancing

From: 27/03/2017 **to** 27/03/2017 **Paid by:** [X] Cash

[ ] Debit/Credit Card

Received by: **Nutech Auto Repair**

Contact us on:
011-5498977 / 0849867342

Code	W/B001
This payment	R250,80
VAT@15%	R xxxx
TOTAL	R xxxx

2.3.1 Describe the type of repairs that was done on John's car. (1)

2.3.2 VAT is charged at 15% in South Africa.

Calculate the VAT amount on this receipt. (3)

2.3.3 Calculate the total amount that John paid for the repairs to his car. (3)

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QUESTION 3

3.1 Roxanne is considering the following two airtime options:

OPTION A	**OPTION B**
Flat rate of R80 per month plus R2,00 per minute.	Prepaid option with no flat rate. All calls will cost R3 per minute.

3.1.1 Study the information below and answer the questions.

OPTION A

The formula for the total cost of Option A is:

Total cost = R80 + R2,00 × total minutes of talk time

Minutes of call time	**A**	10	50	100
Total cost (R)	R80	R100	**B**	R280

TABLE 1

(a) Calculate the value of **A** in TABLE 1 above.

(b) Calculate the value of **B** in TABLE 1 above.

(2 × 2) (4)

3.1.2 Study the information below and answer the questions.

OPTION B

The formula for the total cost of Option B is:

Total cost = R3,00 × number of minutes of talk time

Number of minutes	0	10	50	**D**
Total cost (R)	0	**C**	150	R300

TABLE 2

(a) Calculate the value of **C** in TABLE 2 above.

(b) Calculate the value of **D** in TABLE 2 above.

(2 × 2) (4)

3.1.3 Use TABLE 1 in QUESTION 3.1.1 and TABLE 2 in QUESTION 3.1.2 to draw two straight-line graphs on the grid on ANSWER SHEET 1 (attached).

- Label the graphs as Option A and Option B.
- Give the graphs a suitable heading.
- Label the axes correctly. (13)

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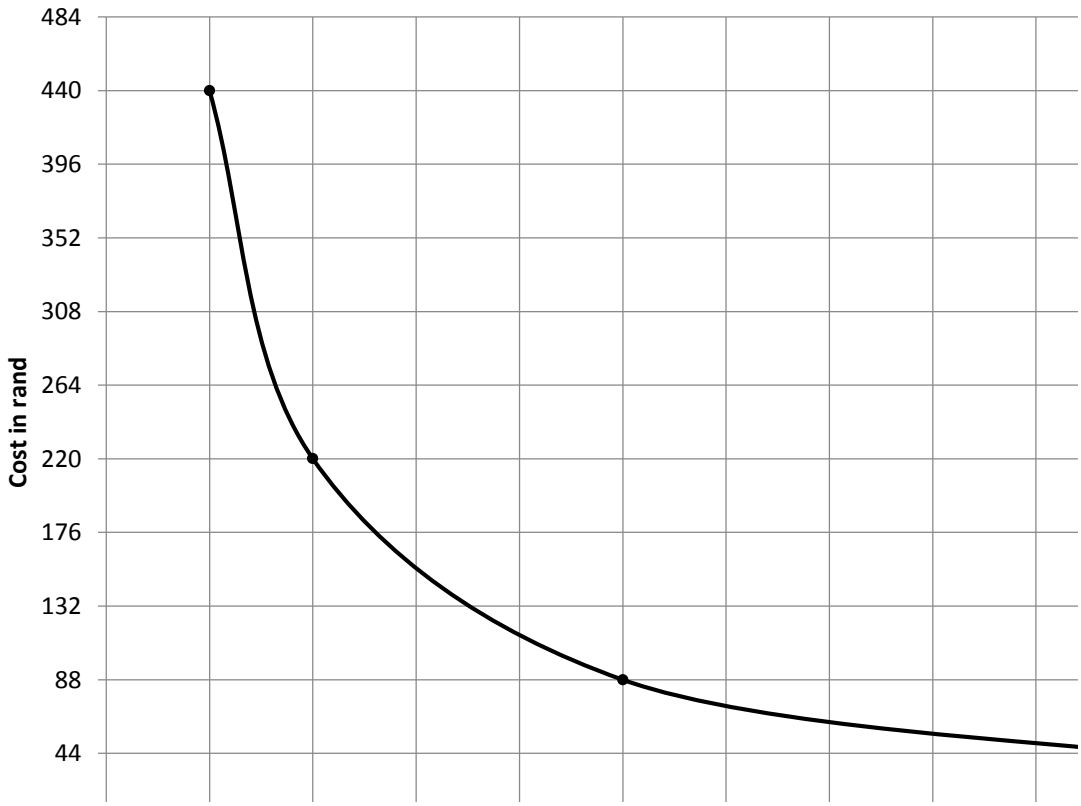
3.1.4 Use your graph to determine the difference in cost between option A and option B if the talk time in a month is 60 minutes. (2)

3.1.5 Use your graph to determine after how many minutes of talk time the cost of Option A and Option B will be the same. (2)

3.2 Study the information below and answer the questions.

Roxanne buys a monthly bus ticket for R440 to travel from home to work daily. The relative cost per trip depends on the total number of trips in a month.

Relative cost per trip = R440,00 ÷ number of trips per month



3.2.1 Describe the above relationship in your own words.

3.2.2 Give the graph a suitable heading.

3.2.3 Determine the relative cost per trip if Roxanne makes 10 trips per month.

3.2.4 How many trips would Roxanne have taken if the relative cost per trip was R88?

3.2.5 Calculate the relative cost per trip if Roxanne makes 25 trips per month.

(5 × 2) (10)

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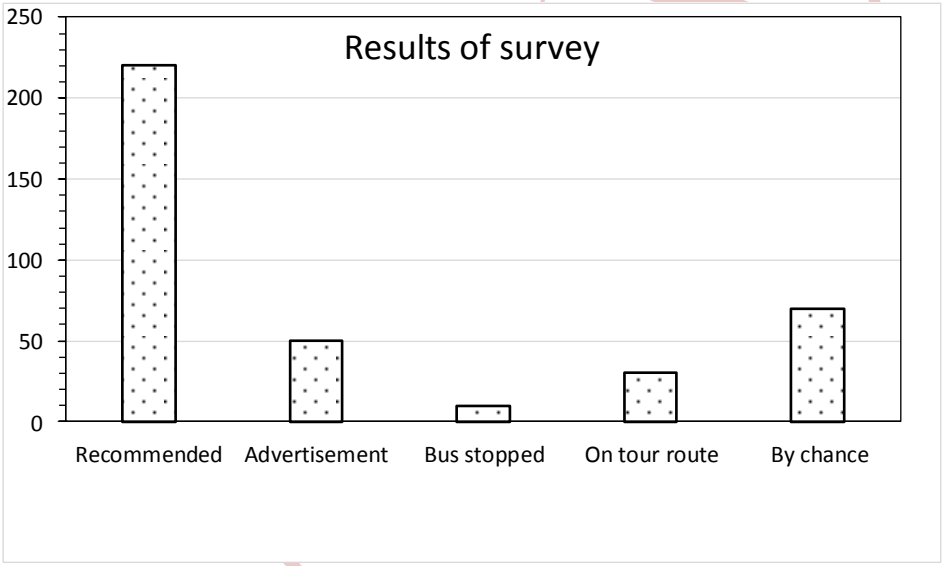
QUESTION 4

4.1 Study the information below and answer the questions.

Maureen owns a traditional decor ornament stall in a tourist area. She used a questionnaire to find out how people heard of her ornament stall. The visitors could choose from the following:

They stopped by chance.	
The stall was on their tour route.	
Their bus stopped there.	
They saw the stall on an advertisement.	
The stall was recommended to them.	

Maureen presented her results of the survey on the bar graph below:



4.1.1 Calculate the total number of visitors that Maureen surveyed. (4)

4.1.2 What was the least frequent reason given and how many of the visitors chose that reason? (2)

4.1.3 Is Maureen's stall popular? Give a reason for your answer. (2)

4.1.4 Calculate the percentage of visitors who stopped at the ornament stall by chance. Round off your answer to two decimals. (3)

4.1.5 Name TWO methods of advertising that Maureen could adopt to increase the number of visitors to her stall. (2)

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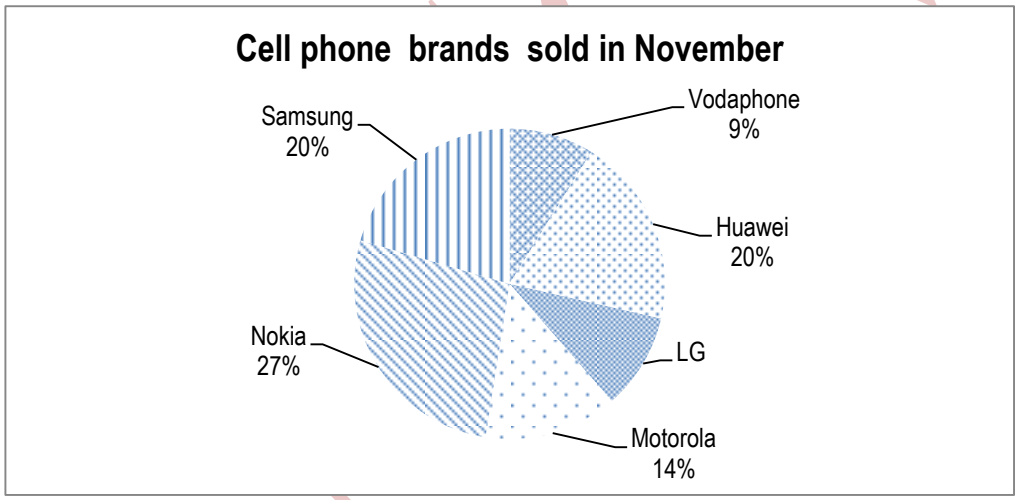
4.2 The table below shows the number of students who use different home languages at a TVET college.

Home language	Number of students
English	75
Pedi	460
Ndebele	185
Tsonga	90

4.2.1 How many students are enrolled at the TVET college? (2)

4.2.2 Use the table above to draw a bar graph on the grid on ANSWER SHEET 2 (attached).
- Give the graph a suitable heading.
- Label the axes correctly. (8)

4.3 Pak cell phone store recorded the cell phone brands that were sold during the month of November. The pie chart below shows the percentages of cell phone brands that were sold in that month.



4.3.1 Which cell phone brand was most popular? (1)

4.3.2 For which cell phone brands was the same percentage of sales recorded? (2)

4.3.3 Determine the percentage sale of LG cell phones. (4)

4.3.4 Pak cell phone shop sold a total of 356 cell phones in November.

Determine the number of Nokia phones that were sold. (3)

4.3.5 Give TWO possible reasons why Vodafone was the cell phone brand that was sold least. (2)

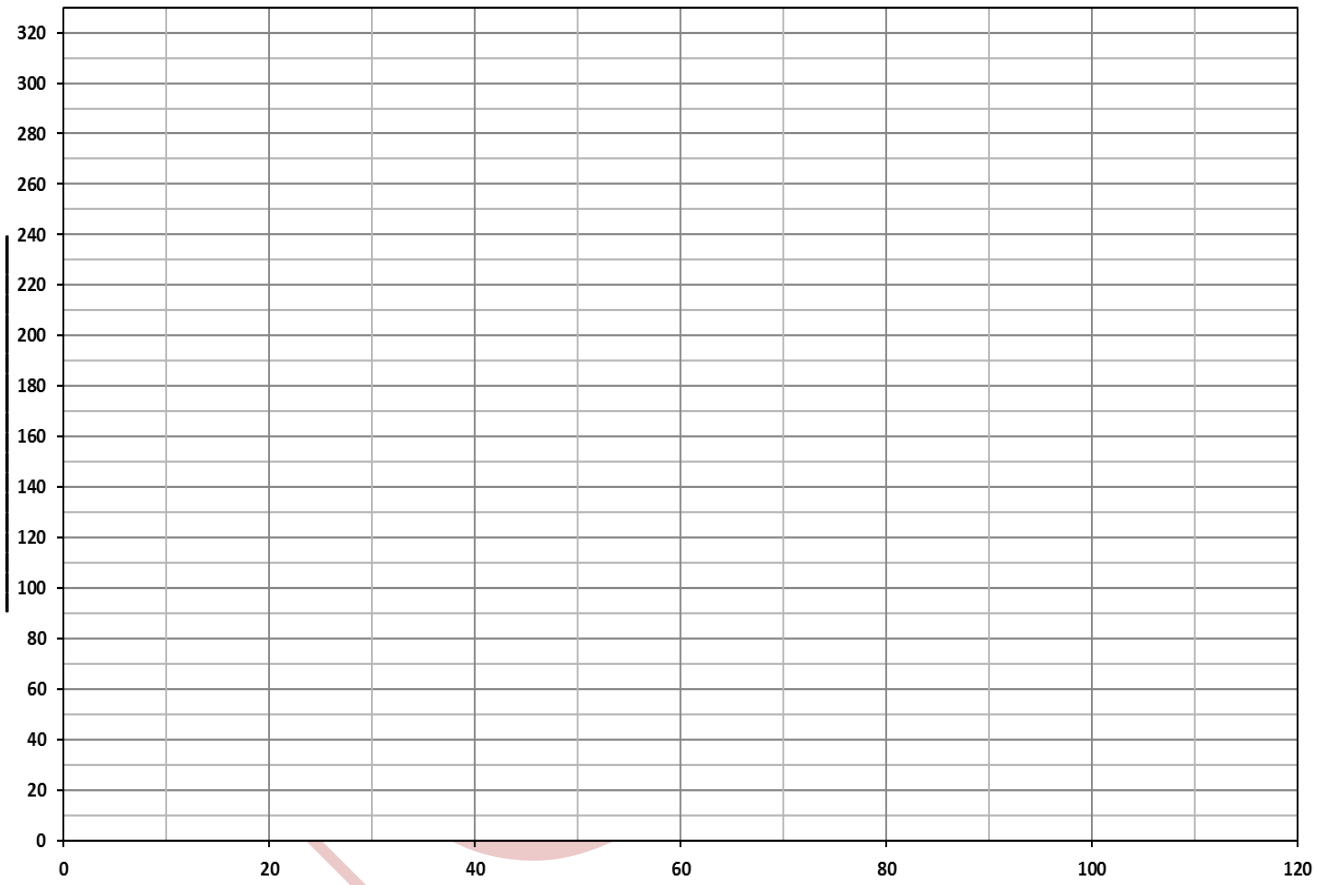
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TOTAL: 150

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ANSWER SHEET EXAMINATION NUMBER:

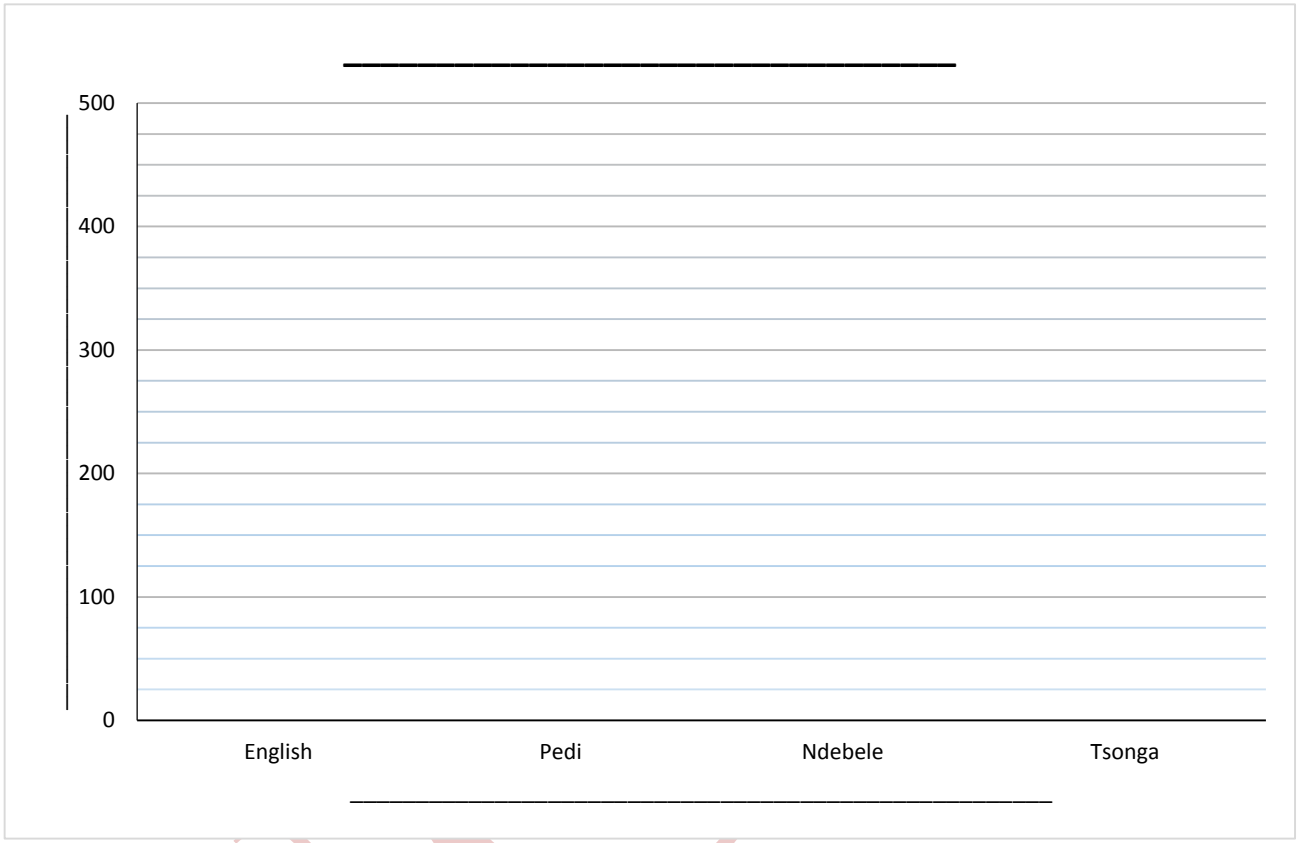
QUESTION 3.1.3



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ANSWER SHEET EXAMINATION NUMBER: | | | | | | | | | | | | | |

QUESTION 4.2.2



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